

PAGE 2026 — {ospsuite} + {esqlabsR} Cheatsheet

Workshop reference — 2 pages

Pavel Balazki · Wilbert de Witte

2026-06-01

{ospsuite} essentials

Workflow

load → inspect → modify → run → extract → analyze
→ plot

Load + save

```
sim <- loadSimulation("path/Model.pkml")
saveSimulation(sim, "path/Model_v2.pkml")
# Independent copy? Reload
sim_copy <- loadSimulation("path/Model.pkml")
```

Inspect

```
getAllParameterPathsIn(sim)
getAllMoleculePathsIn(sim)
getAllContainerPathsIn(sim)
getAllParametersMatching("**|Volume", sim)
```

Parameter access

```
p <- getParameter("Organism|Liver|Volume", sim)
p$value; p$unit
setParameterValuesByPath("Organism|Weight", 75, sim, units =
  ↪ "kg")
```

Run

```
res <- runSimulations(sim)[[1]] # single
res_list <- runSimulations(list(sim1, sim2)) # multiple
```

Extract

```
out <- getOutputValues(res, quantitiesOrPaths =
  ↪ "...|Concentration")
out$data # → data.frame: Time, paths, values
```

Plot — DataCombined

```
dc <- DataCombined$new()
dc$addSimulationResults(res)
dc$addObservedData(loadDataSetFromPKML("obs.pkml"))
plotTimeProfile(dc) # time profile
plotPredictedVsObserved(dc) # GoF
plotResidualsVsCovariate(dc, xAxis = "time") #
  ↪ residuals
```

PK params

```
pk <- calculatePKAnalyses(res, "...|Concentration")
pkAnalysesAsDataFrame(pk) # AUC, C_max, t_max, t½
addUserDefinedPKParameter(...)
```

Local sensitivity

```
sa <- SensitivityAnalysis$new(
  simulation = sim,
  parameterPaths = c("Aciclovir|Lipophilicity",
    "Aciclovir|Fraction unbound (plasma,
      ↪ reference value)"))
sa$addParameterPaths("Organism|Liver|Volume")
sa$numberOfSteps <- 1 # default 2
sa$variationRange <- 0.2 # default 0.1 (symmetric)

saResult <- runSensitivityAnalysis(sa)
saResult$allPKParameterSensitivitiesFor(
  pkParameterName = "C_max",
  outputPath = saResult$allQuantityPaths[[1]],
  totalSensitivityThreshold = 0.9)

exportSensitivityAnalysisResultsToCSV(saResult, "mySA.csv")
```

Units

```
toUnit("Concentration (mass)", 1, "mg/L")
toBaseUnit("Volume", 70, "ml")
toDisplayUnit(p, p$value)
```

Path conventions

Organism|Liver|Volume

Levels: organ → compartment → parameter. | separates levels. ** = recursive wildcard.

Parameter identification

Optimization parameter

```
piPar <- PIParameters$new(
  parameters = getParameter("Aciclovir|Lipophilicity", sim))
piPar$minValue <- -5
piPar$maxValue <- 5
piPar$startValue <- 2 # optional
```

Observed data → output mapping

```
obs <- loadDataSetsFromExcel(
  "Data/Obs.xlsx", createImporterConfigurationForFile(
    filePath = "Data/Obs.xlsx"
  ))

simPath <- "Organism|PeripheralVenousBlood|Aciclovir|Plasma
  ↪ (Peripheral Venous Blood)"
mapping <- PIOutputMapping$new(
  quantity = getQuantity(simPath, container = sim))
mapping$addObservedDataSets(obs$`Aciclovir.Vergin1995.IV250`)
```

Configure + run

```

cfg <- PIConfiguration$new()
cfg$algorithm <- "DEoptim" # or "LevenbergMarquardt"
cfg$autoEstimateCI <- TRUE

piTask <- ParameterIdentification$new(
  simulations = sim,
  parameters = piPar,
  outputMappings = mapping,
  configuration = cfg)

res <- piTask$run()
piTask$plotResults()

```

{esqlabsR} essentials

Project structure

```

MyProject/
├── ProjectConfiguration.xlsx      ← master
├── Models/
│   ├── Simulations/             *.pkml
├── Configurations/
│   ├── Scenarios.xlsx
│   ├── ModelParameters.xlsx
│   ├── Applications.xlsx
│   ├── Individuals.xlsx
│   ├── Populations.xlsx
│   ├── PopulationsCSV/
│   ├── ParameterIdentification.xlsx
│   └── Plots.xlsx
├── Data/                        *.xlsx
├── Results/
└── Reports/                      *.qmd

```

Initialize

```

library(esqlabsR)
initProject(path = "~/MyProject")

```

Project object

```

projectConfig <- esqlabsR::createProjectConfiguration(
  ↪ "workshops/resources/projects/Midazolam-PAGE/ProjectConfiguration"
  ↪ )
print(projectConfig) # full summary

projectConfig$modelFolder #
  ↪ Models/Simulations
projectConfig$configurationsFolder # Configurations/
projectConfig$scenariosFile #
  ↪ Configurations/Scenarios.xlsx
projectConfig$individualsFile #
  ↪ Configurations/Individuals.xlsx
projectConfig$populationsFile #
  ↪ Configurations/Populations.xlsx
projectConfig$parameterIdentificationFile #
  ↪ Configurations/ParameterIdentification.xlsx
projectConfig$plotsFile #
  ↪ Configurations/Plots.xlsx
projectConfig$outputFolder # Results/
projectConfig$dataFile #
  ↪ Data/Midazolam_obs.xlsx

```

Load + run scenarios

```

cfg <- readScenarioConfigurationFromExcel(
  scenarioNames = c("PO_7.5mg", "IV_2mg"),
  projectConfiguration = project)
sc <- createScenarios(cfg)
res <- runScenarios(scenarios = sc)

```

Snapshot / restore

```

snapshotProjectConfiguration(project) # Excel → JSON
  ↪ snapshot
restoreProjectConfiguration(project) # JSON → Excel
projectConfigurationStatus(project) # diff

```

Plots from Excel

Sensitivity (multi-range)

File	One row =	Key columns
Scenarios	one scenario	Scenario, ModelFile, ApplicationProto IndividualId, OutputPathsId
ModelParameters	parameter override	Id, ContainerPath, ParameterName, Value, Units
Applications	dosing protocol	Id, DrugMass, DoseUnit, StartTime, Route
Plots	one figure	PlotID, ScenarioName, OutputPath, xLog, yLog

```
quarto render Reports/MyReport.qmd # default
quarto render Reports/MyReport.qmd -P dose_mg:15 # override
↪ param
quarto render Reports/MyReport.qmd --to pdf # PDF only
```

[illegible]